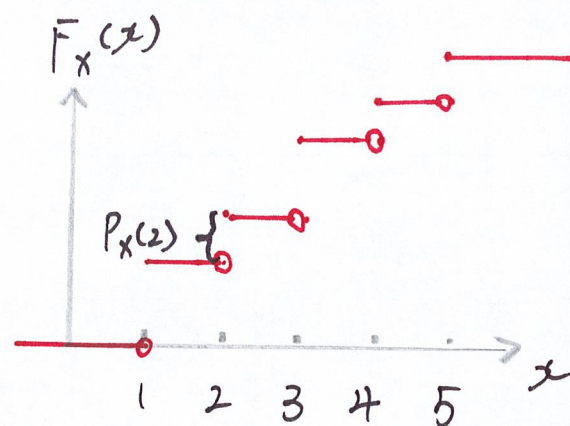


Summary:

▢ For discrete RV's,

$$F_X(x) = \sum_{\forall k \leq x} P_X(k)$$

$$P_X(x) = F_X(x) - F_X(x^-)$$



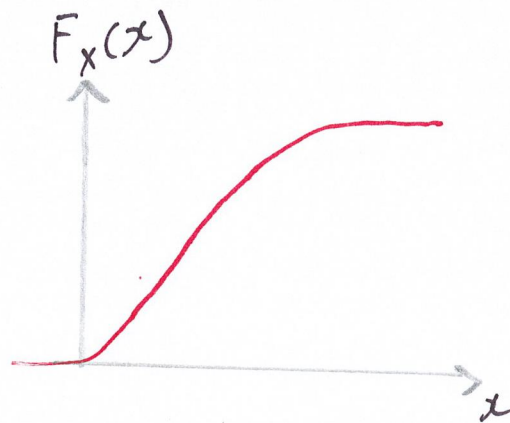
where $x^- = x - \Delta$, $\Delta > 0$ is small.

E.g. $P_X(2) = F_X(2) - F_X(1.999\dots)$

▢ For continuous RV's,

$$F_X(x) = \int_{-\infty}^x f_X(x) dx$$

$$f_X(x) = \frac{d}{dx} F_X(x)$$



$E_x.$

$$F_x(x) = \begin{cases} 0 & , x < 0 \\ \frac{x}{8} & , 0 \leq x < 1 \\ \frac{1}{4} + \frac{x}{8} & , 1 \leq x < 2 \\ \frac{3}{4} + \frac{x}{12} & , 2 \leq x < 3 \\ 1 & , x \geq 3 \end{cases}$$

Find the PMF or PDF of X .

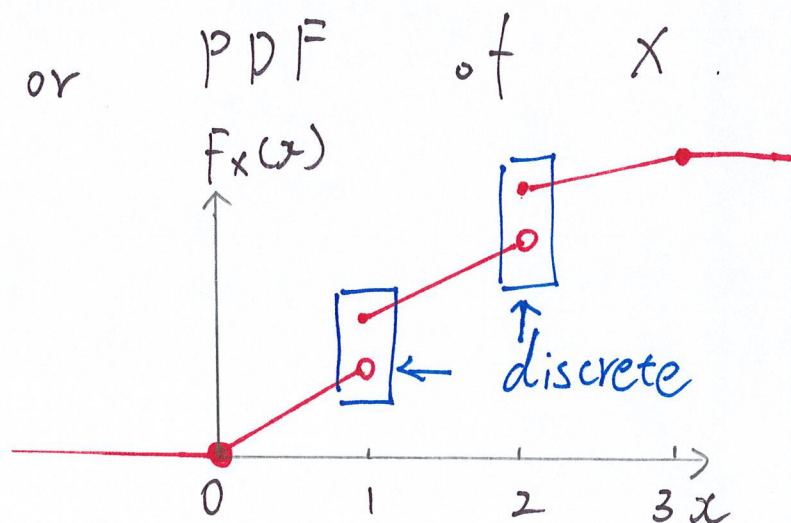
- $F(0^-) = 0$

- $F(0) = 0$

- $F(1^-) = \frac{1}{8}$

- $F(1) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$

- $F(2^-) = \frac{1}{4} + \frac{2}{8} = \frac{4}{8}$



- $F(2) = \frac{3}{4} + \frac{2}{12} = \frac{11}{12}$

- $F(3^-) = \frac{3}{4} + \frac{3}{12} = 1$

- $F(3) = 1$

The dis of X contains two parts :

⇒ Discrete part :

$$P_X(x) = \begin{cases} \frac{2}{8} & , x = 1 \\ \frac{5}{12} & , x = 2 \end{cases}$$

$$F_X(1) - F_X(1^-)$$

$$F_X(2) - F_X(2^-)$$

⇒ Continuous part :

$$f_X(x) = \begin{cases} \frac{1}{8} & , x \in (0, 1) \\ \frac{1}{8} & , x \in (1, 2) \\ \frac{1}{12} & , x \in (2, 3) \end{cases}$$

$$\frac{d}{dx} F_X(x)$$